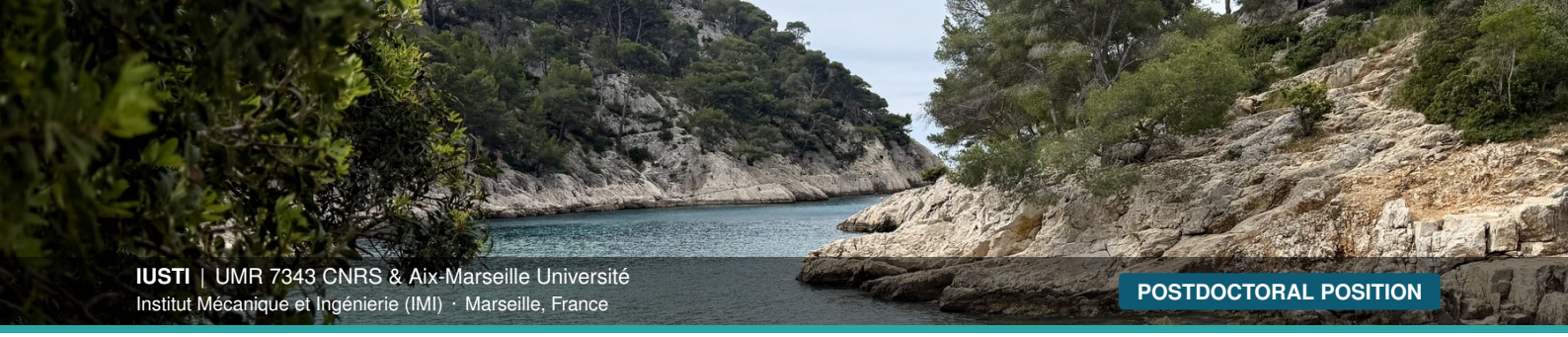


IUSTI | UMR 7343 CNRS & Aix-Marseille Université
Institut Mécanique et Ingénierie (IMI) · Marseille, France

POSTDOCTORAL POSITION

SNSF project *Viscocrack* · Aim 4

Steering cracks with light

Predicting and controlling crack paths in soft solids by digital light patterning

* * *

POSITION	Postdoctoral researcher, experimental	**DURATION**	18 months (project ends August 2028)	**START**	As soon as possible
PARTNERS	UCLA (L. Jin, phase-field modeling) · EPFL (E. Amstad, materials) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. If the path of a crack depends on the local response of the material, then shaping that response in space should allow the path to be designed. Heterogeneous and architected solids already show that spatial variations of stiffness and toughness can deflect, trap, and even rectify cracks [1,2]. Photo-curable soft materials offer a uniquely flexible route to such heterogeneity: with a digital micromirror device, arbitrary grayscale light fields can be projected onto a sample, imprinting the cross-link density, and with it the modulus, relaxation time, and fracture energy, with micrometer resolution [3,4]. This postdoctoral position, dedicated to Aim 4 of *Viscocrack*, will develop a digital light modulation set-up to pattern soft viscoelastic solids, measure how the imprinted landscapes guide crack propagation, and use these experiments to test the predictive power of phase-field simulations developed with the Jin group at UCLA [5]. The long-term goal is inverse design: prescribing a target crack path and computing the light pattern that produces it, and exploring whether patterns updated during propagation can steer a moving crack.

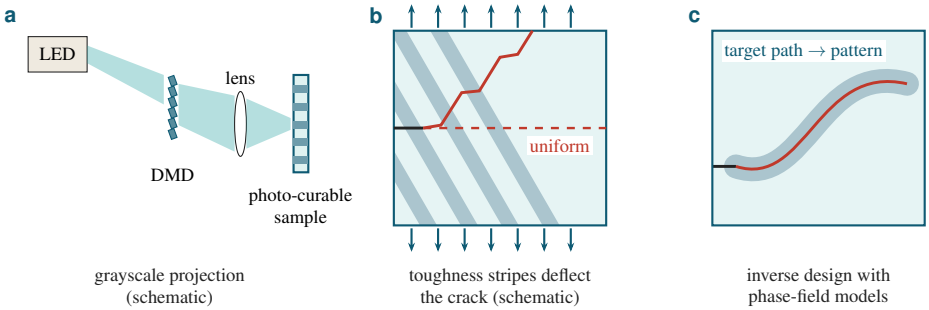


Figure 1. Crack path control by light (schematic). (a) A digital micromirror device (DMD) projects a grayscale light field that sets the local cross-link density of a photo-curable sample. (b) Imprinted stripes of contrasting toughness deflect a crack that would otherwise run straight. (c) Inverse design: phase-field simulations are used to compute the pattern that makes a crack follow a prescribed trajectory.

Context. This position is part of the SNSF project *Viscocrack*, "Crack path prediction and control in nonlinearly viscoelastic materials: in-silico to experiments with viscoelastic and tough hydrogels." Tough hydrogels, elastomers, and soft tissues are increasingly built into medical devices and soft robots, yet we cannot reliably predict where a crack in such a material will go. *Viscocrack* addresses this question through four coordinated aims: materials characterization and Mode I fracture at EPFL (Soft Materials Laboratory, Prof. E. Amstad), phase-field modeling at UCLA (Prof. L. Jin), and experiments on mixed-mode path selection and on crack-path control at Aix-Marseille Université. The experimental work will be carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, whose custom loading, high-speed, and 3D imaging set-ups have moved from EPFL to the Château-Gombert campus. IUSTI and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged. Exchanges and research visits with EPFL and UCLA are funded by the project.

Candidate profile. We seek a candidate with a PhD in physics, mechanics, materials science, or a related field and a strong experimental background. Experience with optics and light patterning (DMD/DLP projection, photolithography, or light-based 3D printing), with photopolymerization of hydrogels or elastomers, or with fracture experiments is a clear asset, as is fluency in quantitative image analysis and instrument control (Python or MATLAB). The postdoc will lead the experimental program of Aim 4 and work day to day with a PhD student on the same aim, so enthusiasm for mentoring and for close exchange between experiments, theory, and simulation is expected. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] M. Z. Hossain, C.-J. Hsueh, B. Bourdin, and K. Bhattacharya, "Effective toughness of heterogeneous media," *J. Mech. Phys. Solids* **71**, 15–32 (2014).
[2] N. R. Brodnik *et al.*, "Fracture diodes: directional asymmetry of fracture toughness," *Phys. Rev. Lett.* **126**, 025503 (2021).
[3] J. Kim, J. A. Hanna, M. Byun, C. D. Santangelo, and R. C. Hayward, "Designing responsive buckled surfaces by halftone gel lithography," *Science* **335**, 1201–1205 (2012).
[4] X. Kuang *et al.*, "Grayscale digital light processing 3D printing for highly functionally graded materials," *Sci. Adv.* **5**, eaav5790 (2019).
[5] C. Miehe, M. Hofacker, and F. Welschinger, "A phase field model for rate-independent crack propagation: robust algorithmic implementation based on operator splits," *Comput. Methods Appl. Mech. Eng.* **199**, 2765–2778 (2010).